

Decent Coaching Classes
VIII CBSE
Mind Your Math - VI — Answer Key

Section A — MCQ

1. (c) 7
2. (c) $a - b + c$
3. (b) $6x^2 - 15x$
4. (b) $2a + 3b$ $[(5a + b) - (3a - 2b) = 5a + b - 3a + 2b = 2a + 3b]$
5. (a) $6xy$
6. (a) $(x + 3)^2$

Section B — VSA

7. Coefficient of $a^2 = 1$; Constant term = 3
8. $(3a + 5b)(5a + 7b) = 15a^2 + 21ab + 25ab + 35b^2 = 15a^2 + 46ab + 35b^2$
9. $15x^2y - 20xy^2 = 5xy(3x - 4y)$
10. $x^2 + 10x + 25 = (x + 5)^2$

Section C — SA (any one option in each)

11. $(a^2 - 3a + 7)(2a + 3) = 2a^3 + 3a^2 - 6a^2 - 9a + 14a + 21 = 2a^3 - 3a^2 + 5a + 21$
OR
 $(a^2 - 5a + 8)(a^2 + 2) = a^4 + 2a^2 - 5a^3 - 10a + 8a^2 + 16 = a^4 - 5a^3 + 10a^2 - 10a + 16$
12. $(2x + 7y)(x - 3y) = 2x^2 - 6xy + 7xy - 21y^2 = 2x^2 + xy - 21y^2$
 $(9x + 4)(3x - 2) = 27x^2 - 18x + 12x - 8 = 27x^2 - 6x - 8$
Difference = $(2x^2 + xy - 21y^2) - (27x^2 - 6x - 8) = -25x^2 + xy - 21y^2 + 6x + 8$
OR
 $(5x^2 - y^2)(x^2 + 3y^2) = 5x^4 + 15x^2y^2 - x^2y^2 - 3y^4 = 5x^4 + 14x^2y^2 - 3y^4$
 $(2xy)^2 = 4x^2y^2$
Sum = $5x^4 + 14x^2y^2 - 3y^4 + 4x^2y^2 = 5x^4 + 18x^2y^2 - 3y^4$
13. $4x^2 + 20x + 25 = (2x)^2 + 2(2x)(5) + 5^2 = (2x + 5)^2$
OR
 $9a^2 - 12ab + 4b^2 = (3a)^2 - 2(3a)(2b) + (2b)^2 = (3a - 2b)^2$

Section D — Case Study

14. (a) $6x^2y + 9xy^2 = 3xy(2x + 3y)$
(b) At $x = 2, y = 1$: Area = $6(2)^2(1) + 9(2)(1)^2 = 24 + 18 = 42$ sq. m
(c) Factorised form : $3xy(2x + 3y) = 3(2)(1)(2 \times 2 + 3 \times 1) = 6 \times 7 = 42$ sq. m — same as (b), hence verified.